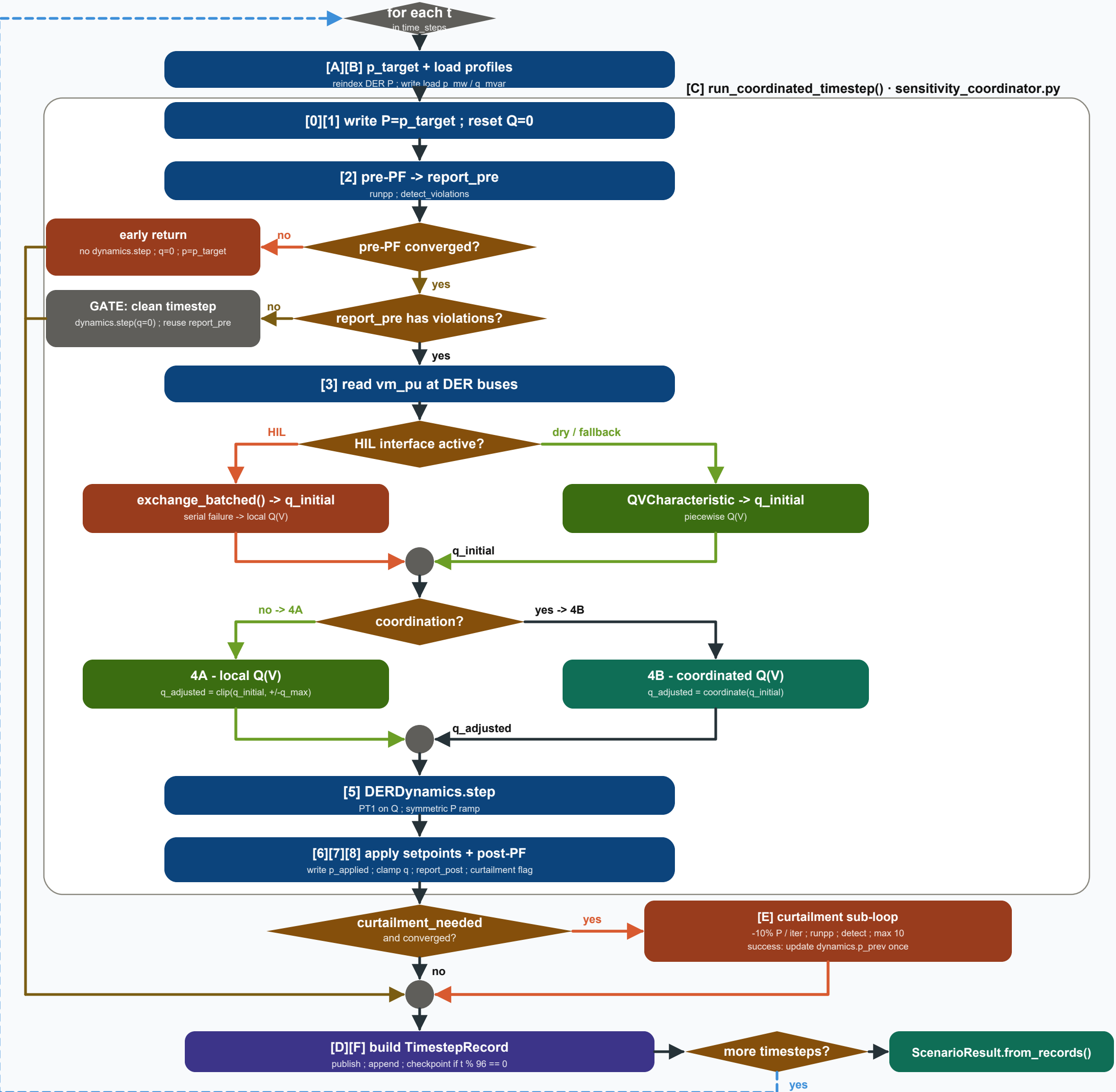




Item 2 - HIL / dry execution

exchange_batched()

DER buses batched to ARDUINO_MAX_DERS
V:<vm> sent at fixed serial precision

Arduino firmware

parse V -> compute q() in float
emit Q:<q> for the same DER order

dry / serial fallback

QVCharacteristic.compute_setpoints()
used for dry-run or serial/protocol failure

Both routes return the same q_initial vector interface

Item 3 - 4A versus 4B

4A - local Q(V)

coordination=False
clip q_initial directly to +/-q_max

4B - coordinated Q(V)

coordination=True
coordinate(q_initial) before dynamics

coordinate() internals

J blocks -> splu(J_PP)
Schur J_red -> residual dV/dQ solve
final q_adjusted clipped to +/-q_max

4A bypasses coordinate(); both modes rejoin at q_adjusted

Items 4-8 - physical response

[5] DERDynamics.step

Q: exact discrete PT1, tau=I95/3
P: symmetric ramp limited by rated power

[6] apply setpoints

write p_applied first
clamp q_applied to network limits

[7][8] post-PF + trigger

runpp -> report_post
residual violations set curtailment_needed

Outer runner - curtailment + record

Curtailment trigger

only after converged post-PF
and residual voltage violations remain

[E] active-P sub-loop

-10% of original P target per iteration
Q is held; no dynamics.step in the sub-loop

[D][F] record + publish

record final post-curtailment/post-Q state
publish -> append -> periodic checkpoint

Execution invariants, state semantics and legend

Execution invariants

1. pre-PF always evaluates p_target with q=0.
2. clean pre-PF skips Q(V), coordinate() and post-PF.
3. failed pre-PF returns before DERDynamics state advances.
4. 4A and 4B share q_initial acquisition and DERDynamics.
5. q is clamped only after p_applied is written to the net.
6. post-PF is the authoritative curtailment trigger.
7. curtailment changes P only; Q remains at the applied value.
8. p_prev is updated once after successful curtailment.

State / branch semantics

q_initial: raw local/HIL Q(V) setpoint
q_adjusted: 4A clip or 4B coordinated setpoint
q_applied / p_applied: dynamic actuator outputs
report_pre: gate decision before Q control
report_post: final Q-control voltage assessment
record: final state after optional active-P curtailment

Legend

- execution
- HIL / curtailment branch
- dry / 4A branch
- gate / record bypass
- next-timestep return